

9. Să se determine coeficienții și nodurile formulelor de cuadratură cu grad maxim de exactitate de forma:

(a)

$$\int_{-1}^1 \frac{f(x)}{\sqrt{1-x^2}} dx = Af(-1) + Bf(x_0) + Cf(1) + R(f).$$

(b)

$$\int_{-1}^1 \frac{f(x)}{\sqrt{1-x^2}} dx = A_0f(x_0) + A_1f(x_1) + A_2f(x_2) + R(f).$$

(c)

$$\int_a^b f(x) dx = Af(a) + Bf\left(\frac{a+b}{2}\right) + Cf(b) + R(f).$$

(d)

$$\int_0^\infty e^{-x} f(x) dx = A_0f(1) + A_1f(x_1) + R(f).$$

Soluție: a)

$$\int_{-1}^1 \frac{f(x)}{\sqrt{1-x^2}} dx = Af(-1) + Bf(x_0) + Cf(1) + R(f)$$

$$R(f) = 0, \text{ pt. } f = e_0, e_1, e_2, e_3$$

$$\begin{aligned} \int_{-1}^1 \frac{dx}{\sqrt{1-x^2}} &= A + B + C \\ \int_{-1}^1 \frac{x dx}{\sqrt{1-x^2}} &= A(-1) + Bx_0 + C \\ \int_{-1}^1 \frac{x^2 dx}{\sqrt{1-x^2}} &= A + Bx_0^2 + C \\ \int_{-1}^1 \frac{x^3 dx}{\sqrt{1-x^2}} &= A(-1) + Bx_0^3 + C \end{aligned}$$

Sistemul în A, B, C, x_0 :

$$\begin{aligned} \pi &= A + B + C, 0 = -A + Bx_0 + C \\ \frac{1}{2}\pi &= A + Bx_0^2 + C, 0 = -A + Bx_0^3 + C \end{aligned}$$

din ec. 2, 4:

$$B(x_0^3 - x_0) = 0$$

rezultă:

$$\begin{aligned} x_0 &= 0 \\ A &= C, A + C = \pi/2 \end{aligned}$$

deci

$$A = C = \pi/4, B = \pi/2$$

Formula:

$$\int_{-1}^1 \frac{f(x)}{\sqrt{1-x^2}} dx = \frac{\pi}{4} (f(-1) + 2f(0) + f(1)) + R(f)$$

$$\begin{aligned} \int_{-1}^1 \frac{x^2 dx}{\sqrt{1-x^2}} &= \int_{-\pi/2}^{\pi/2} \frac{\sin^2 t (\cos t dt)}{\sqrt{1-\sin^2 t}} = \\ &= \int_{-\pi/2}^{\pi/2} \sin^2 t dt = \int_{-\pi/2}^{\pi/2} \frac{1-\cos(2t)}{2} dt = \int_{-\pi/2}^{\pi/2} \frac{1}{2} dt = \pi/2. \end{aligned}$$

b)

$$\int_{-1}^1 \frac{f(x)}{\sqrt{1-x^2}} dx = A_0 f(x_0) + A_1 f(x_1) + A_2 f(x_2) + R(f)$$

x_0, x_1, x_2 sunt rădăcinile pol. lui Cebîșev de grad 3:

$$T_3(x) = \frac{1}{2^2} \cos(3 \arccos x)$$

deci:

$$\begin{aligned} x_k &= -\cos \frac{(2k+1)\pi}{6}, k=0,1,2. \\ x_0 &= -\frac{\sqrt{3}}{2}, x_1=0, x_2=\frac{\sqrt{3}}{2} \end{aligned}$$

Ponderile:

$$\begin{aligned} \int_{-1}^1 \frac{dx}{\sqrt{1-x^2}} &= A_0 + A_1 + A_2 = \pi \\ \int_{-1}^1 \frac{x dx}{\sqrt{1-x^2}} &= -A_0 \frac{\sqrt{3}}{2} + A_2 \frac{\sqrt{3}}{2} = 0 \\ \int_{-1}^1 \frac{x^2 dx}{\sqrt{1-x^2}} &= A_0 \frac{3}{4} + A_2 \frac{3}{4} = \frac{\pi}{2} \end{aligned}$$

rezultă:

$$A_0 \frac{3}{4} 2 = \pi/2, A_0 = \frac{\pi}{3} = A_2 = A_1.$$

c)

$$\int_a^b f(x) dx = Af(a) + Bf\left(\frac{a+b}{2}\right) + Cf(b) + R(f)$$

$A, B, C :$

$$\begin{aligned}\int_a^b dx &= A + B + C = b - a \\ \int_a^b x dx &= Aa + B \frac{a+b}{2} + Cb = \frac{b^2 - a^2}{2} \\ \int_a^b x^2 dx &= Aa^2 + B \left(\frac{a+b}{2} \right)^2 + Cb^2 = \frac{b^3 - a^3}{3}\end{aligned}$$

rezolvând:

$$A = C = \frac{b-a}{6}, B = \frac{2(b-a)}{3}$$

Obs. Făcând calculele rezultă că $R(e_3) = 0, R(e_4) \neq 0$ deci grad de exactitate 3.

d)

$$\int_0^\infty e^{-x} f(x) dx = A_0 f(x_0) + A_1 f(x_1) + R(f)$$

$A_0, A_1, x_0, x_1 :$

$$\begin{aligned}\int_0^\infty e^{-x} 1 dx &= A_0 + A_1 = \Gamma(1) = 1 \\ \int_0^\infty e^{-x} x dx &= A_0 x_0 + A_1 x_1 = \Gamma(2) = 1 \\ \int_0^\infty e^{-x} x^2 dx &= A_0 x_0^2 + A_1 x_1^2 = \Gamma(3) = 2 \\ \int_0^\infty e^{-x} x^3 dx &= A_0 x_0^3 + A_1 x_1^3 = \Gamma(4) = 6\end{aligned}$$

SE știe că:

$$\Gamma(\alpha) = \int_0^\infty e^{-x} x^{\alpha-1} dx, \Gamma(n+1) = n!, n \in \mathbb{N}$$

dar x_0, x_1 sunt rădăcinile polinomului Laguerre de grad 2:

$$\begin{aligned}L_2(x) &= \frac{e^x}{2!} \frac{d^2}{dx^2} (e^{-x} x^2) = \\ &= \frac{e^x}{2!} (-e^{-x} x^2 + 2xe^{-x})' = \\ &= \frac{e^x}{2!} (e^{-x} x^2 - 2xe^{-x} - 2xe^{-x} + 2e^{-x}) = \\ &= \frac{x^2 - 4x + 2}{2}\end{aligned}$$

$x_1 = \sqrt{2} + 2, x_0 = 2 - \sqrt{2}$. A_0, A_1 rezultă:

$$\begin{aligned}A_0 + A_1 &= 1 \\ A_0(2 - \sqrt{2}) + A_1(2 + \sqrt{2}) &= 1\end{aligned}$$

, Solution is: $[A_0 = \frac{1}{4}\sqrt{2} + \frac{1}{2}, A_1 = \frac{1}{2} - \frac{1}{4}\sqrt{2}]$.

$$R(f) = R(e_4) \frac{f^{iv}(\xi)}{4!}$$

(e)

$$\int_1^\infty e^{-2x} f(x) dx = A_0 f(x_0) + A_1 f(x_1) + R(f).$$

(f)

$$\int_{-\infty}^\infty e^{-x^2} f(x) dx = A_0 f(x_0) + A_1 f(x_1) + A_2 f(x_2) + R(f).$$

f) x_0, x_1, x_2 sunt rădăcinile polinomului Hermite de grad 3:

$$\begin{aligned} H_2(x) &= e^{x^2} (e^{-x^2})''' = \\ &= e^{x^2} (-2xe^{-x^2})'' = \\ &= e^{x^2} (-2e^{-x^2} + 4x^2 e^{-x^2}) = \\ &= e^{x^2} (4xe^{-x^2} + 8xe^{-x^2} - 8x^3 e^{-x^2}) = -8x^3 + 12x \end{aligned}$$

$$x_0 = -\sqrt{\frac{3}{2}}, x_1 = 0, x_2 = -x_0.$$

$$\begin{aligned} \int_{-\infty}^\infty e^{-x^2} dx &= \sqrt{\pi} = A_0 + A_1 + A_2 \\ \int_{-\infty}^\infty xe^{-x^2} dx &= 0 = -\sqrt{\frac{3}{2}}A_0 + \sqrt{\frac{3}{2}}A_2 \\ \int_{-\infty}^\infty x^2 e^{-x^2} dx &= \frac{1}{2}\sqrt{\pi} = \frac{3}{2}A_0 + \frac{3A_2}{2} \end{aligned}$$

rezultă:

$$A_0 = A_2 = \frac{\sqrt{\pi}}{3} = A_1$$

(e)

$$\int_1^\infty e^{-2x} f(x) dx = A_0 f(x_0) + A_1 f(x_1) + R(f).$$

Ideea: x_0, x_1 sunt rădăcinile polinomului P_2 de grad 2 ortogonal pe $[1, \infty)$ cu ponderea e^{-2x} , care se determină cu algoritmul Gram-Schmit:

$$P_0(x) = 1$$

$$P_1(x) = x - \alpha P_0(x)$$

unde α :

$$\begin{aligned} \int_1^\infty e^{-2x} P_1(x) P_0(x) dx &= 0 \\ \alpha &= \frac{\int_1^\infty x * 1 e^{-2x} dx}{\int_1^\infty 1^2 e^{-2x} dx} = \frac{\frac{3}{4} e^{-2}}{\frac{1}{2} e^{-2}} = \frac{3}{2} \end{aligned}$$

deci:

$$P_1(x) = x - \frac{3}{2}$$

$$P_2(x) = x^2 - \beta P_1(x) - \gamma P_0(x)$$

$$\begin{aligned} \beta &= \frac{\int_1^\infty e^{-2x} P_1(x) x^2 dx}{\int_1^\infty e^{-2x} P_1^2(x) dx} = \frac{\int_1^\infty e^{-2x} * (x - 3/2) * x^2 dx}{\int_1^\infty e^{-2x} (x - 3/2)^2 dx} = 4 \\ \gamma &= \frac{\int_1^\infty e^{-2x} P_0(x) x^2 dx}{\int_1^\infty e^{-2x} P_0^2(x) dx} = \frac{\int_1^\infty e^{-2x} * x^2 dx}{\int_1^\infty e^{-2x} dx} = \frac{5}{2} \end{aligned}$$

Deci:

$$P_2(x) = x^2 - 4x + \frac{5}{2}$$

Rădăcinile lui P_2 sunt: $x_0 = 2 - \frac{\sqrt{6}}{2}, x_1 = 2 + \frac{\sqrt{6}}{2}$. A_0 și A_1 se determină din sistemul (restul zero pentru e_0 și e_1)

$$\begin{aligned} \int_1^\infty e^{-2x} dx &= A_0 + A_1 \\ \int_1^\infty x * e^{-2x} dx &= \left(2 - \frac{\sqrt{6}}{2}\right) A_0 + \left(2 + \frac{\sqrt{6}}{2}\right) A_1 \end{aligned}$$

rezultă $A_0 = \frac{1}{4} e^{-2} + \frac{1}{24} \sqrt{6} e^{-2}, A_1 = \frac{1}{4} e^{-2} - \frac{1}{24} \sqrt{6} e^{-2}$.